PART 1: Private Registry & Custom Nginx Image

Configuring Docker Insecure Registry (daemon.json)

Docker Engine

v29.2.1

Configure the Docker daemon by typing a json Docker daemon configuration file.

This can prevent Docker from starting. Use at your own risk.

```
{
  "builder": {
    "gc": {
      "defaultKeepStorage": "20GB",
      "enabled": true
    }
  },
  "experimental": false,
  "insecure-registries": [
    "localhost:5000"
  ]
}
```

Running the Local Private Registry Container

```
mohamed@DESKTOP-N92AEE4:~$ docker run -d -p 5000:5000 --restart=always --name registry registry:2
Unable to find image 'registry:2' locally
2: Pulling from library/registry
6d464ea18732: Pull complete
8e82f80af0de: Pull complete
44cf07d57ee4: Pull complete
bbbdd6c6894b: Pull complete
3493bf46cdec: Pull complete
32a76c78501f: Download complete
b537bf6d1146: Download complete
Digest: sha256:a3d8aaa63ed8681a604f1dea0aa03f100d5895b6a58ace528858a7b332415373
Status: Downloaded newer image for registry:2
1e0d92b3aef65c7723a971840a85a2115f6259af98b125d562caedd2c3c88d16
mohamed@DESKTOP-N92AEE4:~$ docker ps
CONTAINER ID   IMAGE        COMMAND                  CREATED          STATUS          PORTS                                       NAMES
1e0d92b3aef6   registry:2   "/entrypoint.sh /etc…"   55 seconds ago   Up 54 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   registry
mohamed@DESKTOP-N92AEE4:~$ mkdir containers/my-nginx-alpine
mohamed@DESKTOP-N92AEE4:~$ cd containers/my-nginx-alpine
mohamed@DESKTOP-N92AEE4:~/containers/my-nginx-alpine$ vim Dockerfile
mohamed@DESKTOP-N92AEE4:~/containers/my-nginx-alpine$
```

Writing the Dockerfile for Alpine-based Nginx

```
FROM alpine:latest
RUN apk add --no-cache nginx
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

Building the Custom Nginx Docker Image

```
mohamed@DESKTOP-N92AEE4:~/containers/my-nginx-alpine$ docker build -t localhost:5000/my-alpine-nginx .
[+] Building 3.0s (6/6) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 131B
 => [internal] load metadata for docker.io/library/alpine:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/2] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
 => [2/2] RUN apk add --no-cache nginx
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:48adf4fb5b813103b5a851c22c861098a45615db6488be0d5833951c755d9292
 => => exporting config sha256:a6788da6f8a1a6b3e84f7c5fe7902a3015548e6cfee1d6c81aa24d9ca1dbd545
 => => exporting attestation manifest sha256:5313acf48690f427a9dd0920b43cd43ba490eb5e791368431bfe67980caa6e52
 => => exporting manifest list sha256:d8ac32544b1dffdff4d175fc30ac18515e9778de8b74ad04e4a497830ede878a
 => => naming to localhost:5000/my-alpine-nginx:latest
 => => unpacking to localhost:5000/my-alpine-nginx:latest
```

Pushing the Image to the Private Registry

```
mohamed@DESKTOP-N92AEE4:~/containers/my-nginx-alpine$ docker push localhost:5000/my-alpine-nginx
Using default tag: latest
The push refers to repository [localhost:5000/my-alpine-nginx]
1dfcbde7ab53: Pushed
931af477c373: Pushed
589002ba0eae: Pushed
latest: digest: sha256:d8ac32544b1dffdff4d175fc30ac18515e9778de8b74ad04e4a497830ede878a size: 855
mohamed@DESKTOP-N92AEE4:~/containers/my-nginx-alpine$
```

Verifying the Pushed Image via Registry API

```
mohamed@DESKTOP-N92AEE4:~/containers/my-nginx-alpine$ curl http://localhost:5000/v2/_catalog
{"repositories":["my-alpine-nginx"]}
mohamed@DESKTOP-N92AEE4:~/containers/my-nginx-alpine$
```

PART 2: WordPress & MySQL with Docker Compose

Docker Compose Configuration for WordPress and MySQL

```
mohamed@DESKTOP-N92AEE4:~/containers$ cd wordpress-mysql
mohamed@DESKTOP-N92AEE4:~/containers/wordpress-mysql$ vim docker-compose.yml
mohamed@DESKTOP-N92AEE4:~/containers/wordpress-mysql$
```

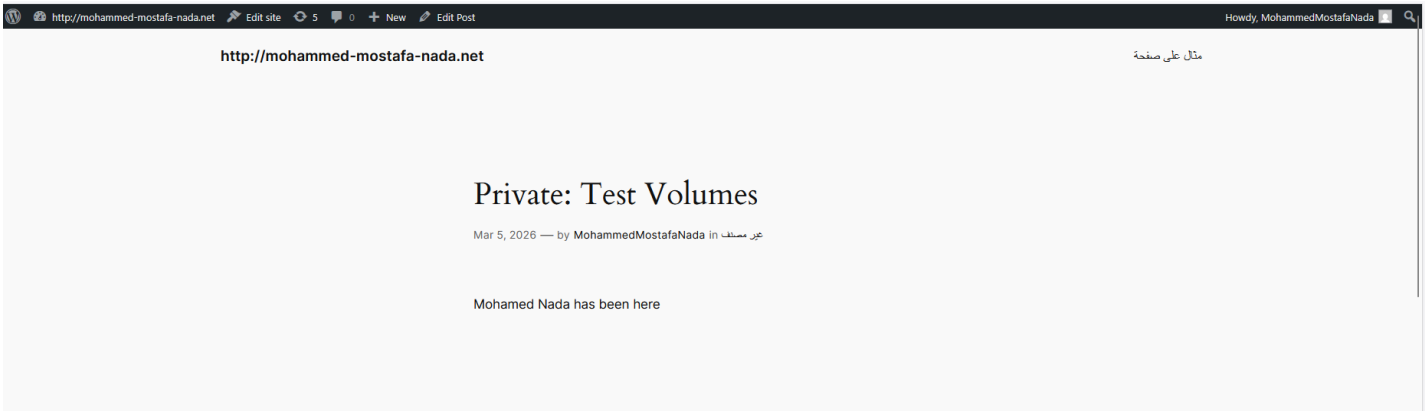
```yaml
services:

  db:
    image: mysql:5.7
    container_name: mysql_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - ./mysql-data:/var/lib/mysql

  wordpress:
    image: wordpress:6.4-apache
    container_name: wordpress_app
    restart: always
    depends_on:
      - db
    ports:
      - "8080:80"
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
```

Starting the WordPress and MySQL Containers

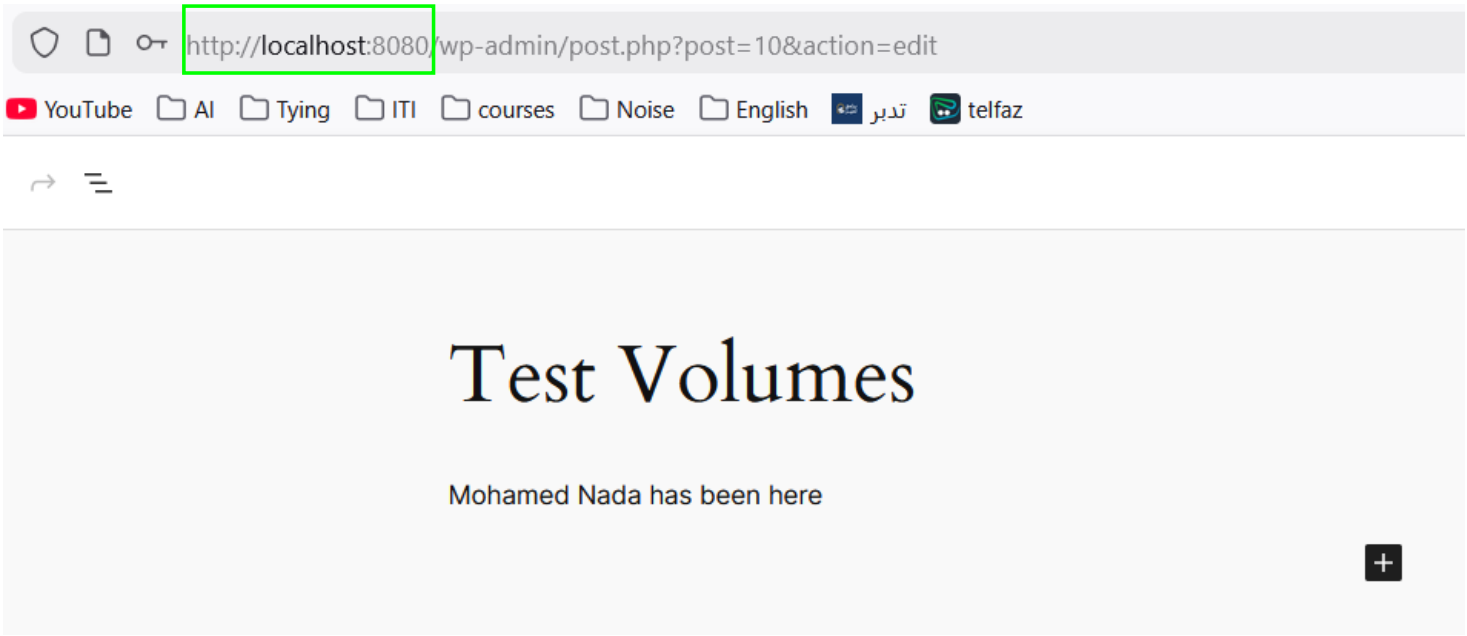
```
mohamed@DESKTOP-N92AEE4:~/containers/wordpress-mysql$ docker compose up -d
WARN[0000] /home/mohamed/containers/wordpress-mysql/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 41/41
 ✓ Image mysql:5.7                Pulled                                                                    113.4s
 ✓ Image wordpress:6.4-apache     Pulled                                                                    133.7s
 ✓ Network wordpress-mysql_default Created                                                                  0.1s
 ✓ Container mysql_db             Created                                                                    0.9s
 ✓ Container wordpress_app        Created                                                                    0.1s
mohamed@DESKTOP-N92AEE4:~/containers/wordpress-mysql$ docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED              STATUS              PORTS                                       NAMES
155650aba4b4   wordpress:6.4-apache  "docker-entrypoint.s…"   About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     wordpress_app
c12837db5984   mysql:5.7            "docker-entrypoint.s…"   About a minute ago   Up About a minute   3306/tcp, 33060/tcp                         mysql_db
1e0d92b3aef6   registry:2           "/entrypoint.sh /etc…"   28 minutes ago       Up 28 minutes       0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp registry
mohamed@DESKTOP-N92AEE4:~/containers/wordpress-mysql$
```

Accessing the WordPress Application on Port 8080

Creating post to test docker volumes




```
mohamed@DESKTOP-N92AEE4:~/containers/wordpress-mysql$ docker compose down
WARN[0000] /home/mohamed/containers/wordpress-mysql/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] down 3/3
 ✓ Container wordpress_app         Removed                                                                                                    1.8s
 ✓ Container mysql_db              Removed                                                                                                    1.5s
 ✓ Network wordpress-mysql_default Removed                                                                                                    0.3s
mohamed@DESKTOP-N92AEE4:~/containers/wordpress-mysql$ docker compose up -d
WARN[0000] /home/mohamed/containers/wordpress-mysql/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 3/3
 ✓ Network wordpress-mysql_default Created                                                                                                    0.1s
 ✓ Container mysql_db              Created                                                                                                    0.1s
 ✓ Container wordpress_app         Created                                                                                                    0.2s
mohamed@DESKTOP-N92AEE4:~/containers/wordpress-mysql$
```



PART 3: BONUS - Flask App & Nginx Reverse Proxy

Pushing the Flask App Image to the Private Registry

```
mohamed@DESKTOP-N92AEE4:~/containers$ docker images

IMAGE                                       ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest                               25109184c71b       13.1MB         3.95MB    U
debian:latest                               3615a749858a        186MB         52.5MB    U
iti-flask-lab2:latest                       19e9c353e5ac       93.1MB         23.1MB    U
localhost:5000/my-alpine-nginx:latest       d8ac32544b1d         16MB         4.74MB
mohammedmostafanada/iti-flask-lab2:latest   19e9c353e5ac       93.1MB         23.1MB    U
mysql:5.7                                   4bc6bc963e6d        700MB          149MB    U
nginx:alpine                                1d13701a5f9f       93.4MB         26.9MB    U
registry:2                                  a3d8aaa63ed8       37.4MB         10.3MB    U
ubuntu:latest                               d1e2e92c075e        119MB         31.7MB    U
wordpress:6.4-apache                        8ae66efb09a2       1.05GB          266MB    U
mohamed@DESKTOP-N92AEE4:~/containers$ docker tag iti-flask-lab2:latest localhost:5000/iti-flask-lab2:latest
mohamed@DESKTOP-N92AEE4:~/containers$ docker push localhost:5000/iti-flask-lab2:latest
The push refers to repository [localhost:5000/iti-flask-lab2]
e965247e4f6f: Pushed
02b83fe93be9: Pushed
3ce7782530be: Pushed
a769d56b7aba: Pushed
be2fee0bf4ec: Pushed
98914020a3be: Pushed
79a000d511eb: Pushed
589002ba0eae: Mounted from my-alpine-nginx
latest: digest: sha256:19e9c353e5aca8e0329fae5f6b5af25f632fd8b3c3367fb2e99b30ed09cc1f38 size: 856
mohamed@DESKTOP-N92AEE4:~/containers$
```

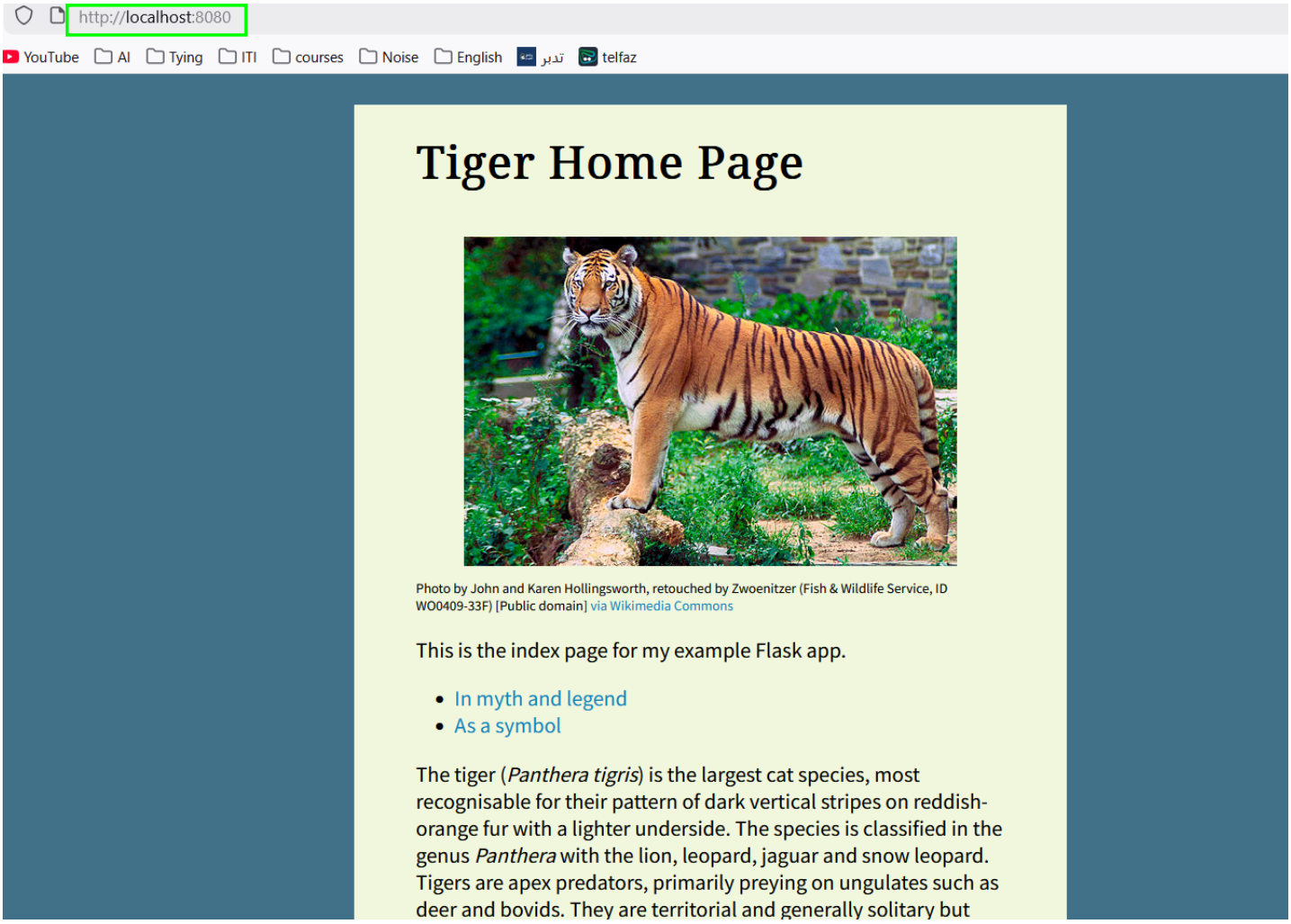
Docker Compose Configuration for Flask App and Nginx Proxy

```
mohamed@DESKTOP-N92AEE4:~/containers$ mkdir ./nginx-bonus
mohamed@DESKTOP-N92AEE4:~/containers$ vim ./nginx-bonus/nginx.conf
mohamed@DESKTOP-N92AEE4:~/containers$ vim docker-compose.yml
```

```
server {
    listen 80;

    location / {
        proxy_pass http://flask_app:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

```
services:
  flask_app:
    image: localhost:5000/iti-flask-lab2
    ports:
      - "8080:5000"

  nginx_proxy:
    image: nginx:alpine
    ports:
      - "80:80"
    volumes:
      - ./nginx.conf:/etc/nginx/conf.d/default.conf:ro
    depends_on:
      - flask_app
```

Starting the Bonus Services Successfully

```
mohamed@DESKTOP-N92AEE4:~/containers/nginx-bonus$ vim docker-compose.yml
mohamed@DESKTOP-N92AEE4:~/containers/nginx-bonus$ docker compose up -d
[+] up 3/3
 ✓ Network nginx-bonus_default         Created                                                    0.1s
 ✓ Container nginx-bonus-flask_app-1   Created                                                    0.1s
 ✓ Container nginx-bonus-nginx_proxy-1 Created                                                    0.1s
mohamed@DESKTOP-N92AEE4:~/containers/nginx-bonus$ docker ps
CONTAINER ID   IMAGE                        COMMAND                  CREATED          STATUS          PORTS                                          NAMES
ffa881b3dc5d   nginx:alpine                 "/docker-entrypoint.…"   5 seconds ago    Up 4 seconds    0.0.0.0:80->80/tcp, [::]:80->80/tcp            nginx-bonus-nginx_proxy-1
ccc39e444439   localhost:5000/iti-flask-lab2 "python routes.py"       5 seconds ago    Up 5 seconds    0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp    nginx-bonus-flask_app-1
1e0d92b3aef6   registry:2                   "/entrypoint.sh /etc…"   About an hour ago Up About an hour 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  registry
mohamed@DESKTOP-N92AEE4:~/containers/nginx-bonus$
```

Testing Application Access via Nginx Proxy (Port 80) and Directly (Port 8080)